

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

103000907 - Time Series Data Mining

DEGREE PROGRAMME

10BA - Master Universitario En Ciencia De Datos

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 2

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1. Description

1.1. Subject details

Name of the subject	103000907 - Time Series Data Mining
No of credits	3 ECTS
Type	Optional/elective
Academic year of the programme	First year
Semester of tuition	Semester 2
Tuition period	February-June
Tuition languages	English
Degree programme	10BA - Master Universitario en Ciencia de Datos
Centre	10 - E.T.S. De Ingenieros Informáticos
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Aurora Perez Perez	D4301	aurora.perez@upm.es	M - 10:30 - 13:30 Th - 10:30 - 13:30
Juan Pedro Caraca-Valente Hernandez (Subject coordinator)	D4301	juanpedro.caracavalente@upm.es	Tu - 09:00 - 12:00 Th - 10:00 - 13:00

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Skills and learning outcomes *

3.1. Skills to be learned

CECD01 - Conocer los procesos de captura, extracción, manipulación y conversión de datos en diferentes entornos.

CECD03 - Manejar las herramientas informáticas para Big Data

CG08 - Capacidad de pensamiento de forma creativa para desarrollar aproximaciones y métodos nuevos y originales

CG11 - Conocimiento y comprensión de la informática para crear modelos, así como sistemas y procesos de información complejos

3.2. Learning outcomes

RA31 - Understand the basics of the Knowledge Discovery Process, and its application to time series and complex data

RA14 - Presentar en público los resultados de sus trabajos de investigación

RA30 - Know how to select and apply the best candidate techniques to Time Series Data Mining projects

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

4. Brief description of the subject and syllabus

4.1. Brief description of the subject

Knowledge Discovery techniques (or Data Mining) in large volumes of information are widely used today in different domains such as medicine, banking environments, industrial systems, etc. with a wide variety of applications such as data analysis, fraud detection, risk analysis, marketing campaigns, etc.

In this course all the stages of the Knowledge Discovery process will be reviewed, focusing on time series, and the most important techniques for each stage will be listed.

In this subject we want to explore areas of Knowledge Discovery less known, but are becoming increasingly relevant. There are domains where information is presented mostly in the form of Time Series which require a very specialized treatment. Examples of these are medical domains such as Electrocardiography or Audiometry, financial domains, etc. Time series are a challenge to the traditional techniques of Data Mining and often require the use of novel solutions. We will discuss traditional numeric Time Series Techniques, novel approaches and will pay special attention to Symbolic approaches.

4.2. Syllabus

1. Introduction

1.1. Time Series

1.2. Basic Concepts

2. Numeric Techniques: Classic Approach

2.1. Introduction to Time Series Distances

2.2. Fourier Transform Approach

2.3. Pattern Discovery

3. Numeric Techniques: Forecasting

3.1. Introduction to Autocorrelation

3.2. Basic ARIMA Models

3.3. Seasonal ARIMA Models

3.4. Multivariate ARIMA Models

3.5. Other Forecasting Techniques

4. Symbolic Techniques for Time Series

4.1. Domain Independent Symbolic Techniques

4.2. Domain Dependent Temporal Abstraction

4.3. Case Studies

5. Schedule

5.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	1. Introduction Duration: 01:00 Lecture 2.Numeric Techniques: Introduction Duration: 01:00 Lecture Time Series Generation: Understanding the underlying components of a real world time series Duration: 01:00 Additional activities			Time Series Generation: Understanding the underlying components of a real world time series Individual work Progressive assessment Presential Duration: 01:00
2	2.1 Time Series Distances Duration: 00:45 Lecture 2.2 Fourier Transform Duration: 00:30 Lecture 2.3 Pattern Discovery Duration: 00:45 Lecture Times Series Clustering Duration: 01:00 Additional activities			Time Series Clustering Individual work Progressive assessment Presential Duration: 01:00
3	Domain Analysis and KDD Process Duration: 00:25 Cooperative activities 3.1 Introduction to Autocorrelation Duration: 00:20 Lecture 3.2 Basic ARIMA Models Duration: 01:15 Lecture Applying Basis ARIMA Models with R Duration: 01:00 Additional activities			Project Stage 1: Domain Analysys, Data study, Objective definition Individual work Progressive assessment Not Presential Duration: 00:20 Applying Basic ARIMA Models with R Individual work Progressive assessment Presential Duration: 01:00

4	3.3 Seasonal ARIMA Models Duration: 01:00 Lecture 3.4 Multivariate ARIMA Models Duration: 01:00 Lecture Applying Seasonal and Multivariate ARIMA Models with R Duration: 01:00 Additional activities			Applying Seasonal and Multivariate ARIMA Models with R Individual work Progressive assessment Presential Duration: 01:00
5	Group Discussion Duration: 01:00 Additional activities Techniques for Time Series Forecasting Duration: 02:00 Inverted classroom			Techniques for Time Series Forecasting Individual presentation Progressive assessment Presential Duration: 02:00
6	4.1 Symbolic Techniques for Time Series: Domain Independent Duration: 02:00 Lecture Applying SAX or other Symbolic Technique Duration: 01:00 Additional activities			Applying SAX or other Symbolic Technique Individual work Progressive assessment Presential Duration: 01:00
7	4.2 Symbolic techniques: Temporal Abstraction Duration: 00:30 Lecture Case Study: Temporal Abstraction Duration: 01:30 Problem-solving class Applying Temporal Abstraction Duration: 01:00 Additional activities			Project Stage 2: Application of Data Mining Techniques Individual work Progressive assessment Not Presential Duration: 00:20 Applying Temporal Abstraction Group work Progressive assessment Presential Duration: 01:00
8	Project Presentations Duration: 03:00 Additional activities			Project Presentations Individual presentation Progressive assessment Presential Duration: 03:00
9				
10				
11				
12				
13				
14				
15				
16				

17				Project Delivery Individual work Global examination Not Presential Duration: 02:00
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Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

6. Activities and assessment criteria

6.1. Assessment activities

6.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
1	Time Series Generation: Understanding the underlying components of a real world time series	Individual work	Face-to-face	01:00	5%	/ 10	CG11 CECD01 CECD03
2	Time Series Clustering	Individual work	Face-to-face	01:00	5%	/ 10	CG08 CG11 CECD01 CECD03
3	Project Stage 1: Domain Analysis, Data study, Objective definition	Individual work	No Presential	00:20	10%	/ 10	CECD01 CECD03
3	Applying Basic ARIMA Models with R	Individual work	Face-to-face	01:00	5%	/ 10	CG08 CG11 CECD01 CECD03
4	Applying Seasonal and Multivariate ARIMA Models with R	Individual work	Face-to-face	01:00	5%	/ 10	CG08 CG11 CECD01 CECD03
5	Techniques for Time Series Forecasting	Individual presentation	Face-to-face	02:00	10%	/ 10	CG08 CG11 CECD01 CECD03
6	Applying SAX or other Symbolic Technique	Individual work	Face-to-face	01:00	5%	/ 10	CG08 CG11 CECD01 CECD03
7	Project Stage 2: Application of Data Mining Techniques	Individual work	No Presential	00:20	10%	/ 10	CG08 CG11 CECD03
7	Applying Temporal Abstraction	Group work	Face-to-face	01:00	5%	/ 10	CG08 CG11 CECD01 CECD03

8	Project Presentations	Individual presentation	Face-to-face	03:00	50%	/ 10	CG08 CG11 CECD01 CECD03
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6.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Project Delivery	Individual work	No Presential	02:00	100%	5 / 10	CG08 CG11 CECD01 CECD03

6.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Project complete	Group work	Face-to-face	00:00	100%	5 / 10	CG08 CG11 CECD01 CECD03

6.2. Assessment criteria

For the evaluation of this subject we will take into account, on the one hand, the Progressive Evaluation Activities and, on the other hand, the Data Mining Project.

For the **Progressive Evaluation** we will consider:

- the attendance to class
- the active participation of the student
- the tasks that will be done in each class applying the concepts explained previously. This will be the most important part of the Progressive Evaluation

As they are carried out during the course, progressive evaluation activities are only evaluable for the

progressive evaluation, and can not be repeated for the Final Evaluation or the extra call.

The **Data Mining Project** will be evaluated according to the phases described below and their corresponding weights.

This project will be done individually or in groups of 2 people. The work will be done incrementally and will be presented in the following phases:

- Phase 1: students will choose a domain to which data they have access, analyze their characteristics and establish the objectives to be achieved through the Data Mining Project. They will write a report describing the domain they have selected, analysing the data and making an initial proposal of the objectives of the project.
- Phase 2: through the use of a Knowledge Discovery software tool or just a programming language, Data Mining algorithms will be applied to the data of each domain. The results should be evaluated and analyzed. In addition, the student will analyze the limitations of the algorithms used and possible improvements.

The two deliveries of the Data Mining Project are mandatory and will be evaluated according to the weights assigned in the table in the previous section (summative evaluation).

The Data Mining Project will be presented in class. Each group will have approximately 15 minutes for the oral presentation plus 5 minutes of questions.

Qualification standards

- Progressive Evaluation: The subject will be evaluated on 10 points, divided into 4 points for Progressive Evaluation Activities and 7 for the Data Mining Project (yes, more than 10 points are available to encourage consistency in doing the daily activities). The dates for the delivery of each part of the Data Mining Project will be published at the beginning of the course.

- Final Evaluation and the extra call: Only the Data Mining Project will be considered. Those parts of the Data Mining Project that are pending may be delivered. Continuous assessment will not be repeated, but will be taken into account if done during the course.

7. Teaching resources

7.1. Teaching resources for the subject

Name	Type	Notes
Time Series Analysis and Its Applications With R Examples - 4th Edition	Bibliography	Book on Time Series Analysis by Shumway and Stoffer
A review on time series data mining	Bibliography	Engineering Applications of Artificial Intelligence 24 (2011) 164?181
Data Mining: Concepts and Techniques	Bibliography	Book about Data Mining Techniques. J.Han y M. Kamber. Ed. Morgan Kauffman, 2006.
Data Mining: Concepts, Models, Methods, and Algorithms	Bibliography	Book about Data Mining Techniques. M. Kantardzic (eds.), John Wiley & Sons, 2003
From Data Mining to Knowledge Discovery in Databases	Bibliography	Book about some foundational works on nowadays Data Mining Techniques U. Fayyad, G. Piatetsky-Shapiro y P. Smyth, 1996
WEKA	Web resource	Official webpage of the Data Mining Tool WEKA, with tutorials and free download http://www.cs.waikato.ac.nz/ml/weka/

Moodle	Others	https://moodle.upm.es/titulaciones/oficiales/course/view.php?id=407
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8. Other information

8.1. Other information about the subject

During the course, we will try to use as many the data files related to Sustainable Development Goals of UN as possible, specially number 13 Climate Action